

Bags (Multisets)

Example 1 : Bag Type

Bags are unordered collections that allow duplicates:

bag $\mathbb{N}$
bag $\mathbb{Z}$

Example 2 : Bag Literals

Bags are written with double square brackets:

$\llbracket x \rrbracket$
 $\llbracket a, b, c \rrbracket$
 $\llbracket 1, 2, 2, 3, 3, 3 \rrbracket$

Example 3 : Bags vs Sets

Unlike sets, bags preserve multiplicity:

The bag $\llbracket 1, 2, 2, 3 \rrbracket$ is different from the set $\{1, 2, 3\}$

The bag $\llbracket a, a, a \rrbracket$ contains three copies of a

Example 4 : Bags elem Specifications

Bags can model collections where order doesn't matter but quantity does:

$coins \in \text{bag } Coin$
 $items \in \text{bag } Item$